

Document 14 – Executive Dashboard Design

TUI Smart Destination Recommender

Executive Dashboard Design

Version

1.0

Purpose

This document defines the Executive Dashboard strategy for the TUI Smart Destination Recommender.

The dashboard provides business stakeholders with a centralized view of tourism demand, sustainability performance, recommendation effectiveness, and congestion management outcomes.

The objective is to transform recommendation outputs into actionable business intelligence that supports strategic decision-making and responsible tourism initiatives.

1. Dashboard Objectives

The Executive Dashboard supports four primary objectives.

Objective 1 – Business Visibility

Provide a comprehensive view of recommendation system performance.

Objective 2 – Sustainability Monitoring

Measure how recommendations support sustainable tourism goals.

Objective 3 – Congestion Management

Monitor visitor distribution and overtourism risks.

Objective 4 – Decision Support

Enable business users to make informed operational and strategic decisions.

2. Target Users

The dashboard serves multiple stakeholder groups.

Executive Leadership

Focus Areas:

- Strategic KPIs
 - Sustainability performance
 - Tourism demand trends
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Product Managers

Focus Areas:

- Recommendation effectiveness
 - User engagement
 - Destination performance
-

Sustainability Teams

Focus Areas:

- Sustainability metrics
 - Environmental impact indicators
 - Sustainable tourism adoption
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Operations Teams

Focus Areas:

- Congestion monitoring
 - Capacity management
 - Destination balancing
-

3. Dashboard Architecture

Data Sources



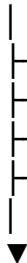
BigQuery



Data Mart Layer



Executive Dashboard



Business KPIs

Sustainability KPIs

Congestion KPIs

Recommendation KPIs

Business Users

The architecture separates analytical reporting from operational systems.

4. Dashboard Sections

The dashboard consists of four primary modules.

Module 1 – Executive Overview

Provides a high-level summary of platform performance.

Module 2 – Recommendation Analytics

Measures recommendation effectiveness.

Module 3 – Sustainability Monitoring

Tracks environmental objectives.

Module 4 – Congestion Management

Monitors tourism distribution.

5. Executive Overview Page

Purpose:

Provide a one-page summary of business performance.

Key Metrics

Total Destinations

Example:

150

Active Users

Example:

25,000

Total Recommendations Generated

Example:

1.2 Million

Recommendation Acceptance Rate

Example:

42%

Average Sustainability Score

Example:

84/100

Average Congestion Score

Example:

58/100

Executive Overview Layout

| Total Users | Recommendations | Acceptance % |

| Sustainability | Congestion | Destinations |

Recommendation Trends Over Time

6. Recommendation Analytics Dashboard

Purpose:

Measure recommendation engine performance.

KPIs

Recommendation Volume

Number of recommendations generated.

Top Recommended Destinations

Most frequently recommended destinations.

Recommendation Acceptance Rate

Percentage of recommendations resulting in bookings.

Average Recommendation Score

Average ranking score generated by the engine.

Recommendation Confidence

Average recommendation confidence score.

Visualizations

Top Recommended Destinations

Bar Chart

Recommendation Trends

Line Chart

Recommendation Distribution

Histogram

Confidence Score Distribution

Box Plot

7. Sustainability Dashboard

Purpose:

Track environmental performance.

KPIs

Average Sustainability Score

Measures overall recommendation sustainability.

Sustainable Destination Ratio

Percentage of recommendations directed toward highly sustainable destinations.

Sustainability Impact Score

Measures positive environmental contribution.

Green Destination Adoption

Percentage of bookings generated from sustainable destinations.

Visualizations

Sustainability Score Distribution

Histogram

Sustainable vs Non-Sustainable Recommendations

Pie Chart

Sustainability Trend

Line Chart

Top Sustainable Destinations

Sustainability Dashboard Layout

| Avg Sustainability | Green Adoption Rate |

| Sustainability Trend |

| Sustainable Destinations Ranking |

8. Congestion Management Dashboard

Purpose:

Measure overtourism reduction effectiveness.

KPIs

Average Congestion Score

Measures overall congestion exposure.

High Congestion Destination Ratio

Percentage of recommendations directed to highly congested locations.

Diversification Rate

Measures recommendation distribution across destinations.

Congestion Reduction Impact

Estimated reduction in overcrowded destination exposure.

Visualizations

Congestion Score Distribution

Histogram

High-Risk Destinations

Heat Map

Destination Diversification

Treemap

Congestion Trends

Line Chart

Congestion Dashboard Layout

Avg Congestion Diversification Rate	

Congestion Trend	

Destination Congestion Heatmap	

9. Explainable AI Dashboard

Purpose:

Provide visibility into recommendation logic.

KPIs

Recommendation Confidence

Average confidence score.

Sustainability Influence

Percentage of rankings impacted by sustainability optimization.

Congestion Influence

Percentage of rankings adjusted by congestion balancing.

Top Recommendation Drivers

Most influential recommendation factors.

Visualizations

Feature Contribution Analysis

Bar Chart

Recommendation Drivers

Ranking Table

Confidence Distribution

Histogram

Recommendation Explanation Examples

Interactive Panel

10. Destination Insights Dashboard

Purpose:

Provide detailed destination-level analytics.

Metrics

Destination Popularity

Sustainability Score

Congestion Score

Recommendation Frequency

Booking Conversion Rate

Example Destination Profile

Destination:

Madeira

Sustainability Score:

91

Congestion Score:

34

Recommendation Frequency:

8.2%

Booking Conversion:

18%

11. Filters and Interactivity

Users can filter dashboard content by:

Region

Examples:

- Europe
 - Asia
 - Americas
-

Destination Type

Examples:

- Beach
 - Nature
 - Urban
 - Cultural
-

Sustainability Level

Examples:

- High
 - Medium
 - Low
-

Time Period

Examples:

- Last 30 Days
 - Last Quarter
 - Last Year
-

12. Technology Stack

Recommended implementation options.

Option 1 – Looker Studio

Benefits:

- Native Google Cloud integration
- Rapid development
- Executive-friendly interface

Recommended for MVP.

Option 2 – Power BI

Benefits:

- Enterprise reporting
 - Advanced visualizations
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Option 3 – Streamlit

Benefits:

- Full customization
- AI-focused experiences
- Interactive demonstrations

Recommended for hackathon demonstrations.

13. Future Dashboard Enhancements

Future capabilities may include:

Real-Time Monitoring

Live tourism congestion tracking.

Predictive Analytics

Tourism demand forecasting.

Sustainability Forecasting

Environmental impact prediction.

AI Assistant

Natural language dashboard exploration.

Example:

"Show destinations with sustainability scores above 85 and congestion below 40."

14. Business Value

The Executive Dashboard provides value across multiple dimensions.

For Executives

- Strategic visibility
 - Sustainability reporting
 - Performance monitoring
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For Product Teams

- Recommendation optimization
 - User behavior analysis
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For Sustainability Teams

- Environmental impact monitoring
 - Sustainable tourism measurement
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For Operations Teams

- Congestion management

- Capacity planning
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Conclusion

The Executive Dashboard transforms recommendation outputs into actionable business intelligence. By combining recommendation analytics, sustainability monitoring, congestion management, explainable AI insights, and destination intelligence, the dashboard provides a comprehensive decision-support platform aligned with TUI's strategic objectives and responsible tourism vision.